

2.2 Simplification and Substitution

In the previous section we introduced variables, terms, coefficients and the basic techniques of algebra. In this section we combine those skills to simplify larger expressions, factorise expressions and evaluate formulae by substitution. These are essential skills throughout engineering mathematics.

2.2.1 Simplifying Expressions

Simplifying an algebraic expression often requires several steps. Typical operations include:

- Expanding brackets.
- Collecting like terms.
- Simplifying numerical coefficients.
- Factorising.

The aim is to write the expression in its simplest form without changing its value.

Worked Example

Simplify

$$3(x + 4) + 2x - 5$$

First expand the bracket:

$$3x + 12 + 2x - 5$$

Now collect like terms and simplify:

$$\begin{aligned} 3x + 12 + 2x - 5 &= 3x + 2x + 12 - 5 \\ &= 5x + 7 \end{aligned}$$

Worked Example

Simplify

$$4(2y - 3) + 5(6 - 3y)$$

$$4(2y - 3) + 5(6 - 3y) = 8y - 12 + 30 - 15y \quad \text{expanding brackets}$$

$$= 8y - 15y - 12 + 30 \quad \text{collecting like terms}$$

$$= 18 - 7y \quad \text{simplifying}$$

2.2.2 Negative Brackets

Negative brackets are a common source of errors. A minus sign immediately before a bracket means that every term in the bracket is multiplied by -1 .

Worked Example

Simplify

$$-(x + 5)$$

Multiplying every term by -1 gives

$$-x - 5$$

Worked Example

Simplify

$$4x - (2x - 3)$$

Remove the bracket carefully:

$$4x - 2x + 3$$

Now collect like terms and simplify:

$$2x + 3$$

Therefore,

$$4x - (2x - 3) = 2x + 3$$

Worked Example

Simplify

$$5a - (3a + 7) + 2$$

$$\begin{aligned} 5a - (3a + 7) + 2 &= 5a - 3a - 7 + 2 && \text{multiply out the brackets} \\ &= 2a - 5 && \text{collect like terms and simplify} \end{aligned}$$

2.2.3 Factorisation

Expanding brackets and factorisation are opposite processes. For example,

$$3x(x + 2) = 3x^2 + 6x$$

Factorisation starts with the expanded form and works backwards.

Definition

Factorisation is the process of identifying common factors and writing them outside a bracket.

Worked Example

Factorise

$$3x + 6$$

Both terms contain a factor of 3. Removing the common factor gives

$$3(x + 2)$$

Worked Example

Factorise

$$3x + 2xy$$

Both terms contain x . Removing x gives

$$x(3 + 2y)$$

Worked Example

Factorise

$$12x^2 - 8xy^2$$

The largest common factor is $4x$. Removing $4x$ gives

$$12x^2 - 8xy^2 = 4x(3x - 2y^2)$$

2.2.4 Simple Algebraic Fractions

Before studying algebraic fractions in detail, it is useful to see that many algebraic fractions can be simplified exactly like numerical fractions.

Worked Example

Simplify

$$\frac{6x}{3}$$

Dividing the numerator and denominator by 3 gives

$$2x$$

Worked Example

Simplify

$$\frac{12x^2}{4x}$$

Simplify the numerical part:

$$\frac{12}{4} = 3$$

Simplify the algebraic part (using rules of indices):

$$\frac{x^2}{x} = x$$

Therefore,

$$\frac{12x^2}{4x} = 3x$$

Notice that both numbers and variables can be cancelled provided they appear as factors in both the numerator and denominator.

2.2.5 Substitution

Substitution means replacing a variable with a numerical value.

Definition

To substitute a value into an expression:

1. Replace the variable with its value.
2. Use brackets when substituting.
3. Evaluate using BIDMAS.

Worked Example

Given $x = 4$ evaluate $3x + 2$.

Substituting $x = 4$ gives:

$$\begin{aligned} 3(4) + 2 &= 12 + 2 \\ &= 14 \end{aligned}$$

Worked Example

Given $a = 3$, evaluate $2a^2 - 5$.

Substituting $a = 3$ gives:

$$\begin{aligned} 2(3)^2 - 5 &= 2(9) - 5 && \text{applying BIDMAS rules} \\ &= 18 - 5 \\ &= 13 \end{aligned}$$

2.2.6 Substituting Multiple Variables

Often an expression contains more than one variable.

Worked Example

Given $a = 2$ and $b = 5$, evaluate $3a + b$.

Substituting the values gives

$$\begin{aligned} 3(2) + 5 &= 6 + 5 \\ &= 11 \end{aligned}$$

Worked Example

Given $a = 3$ and $b = 2$, evaluate $2a^2 - 3b$.

Substituting carefully gives:

$$\begin{aligned} 2(3)^2 - 3(2) &= 2(9) - 6 && \text{applying BIDMAS} \\ &= 18 - 6 \\ &= 12 \end{aligned}$$

2.2.7 Substituting Negative Numbers

When substituting negative values, brackets are essential.

Worked Example

Given $x = -2$, evaluate x^2 .

Substituting carefully gives:

$$(-2)^2 = 4$$

Worked Example

Given $x = -5$, evaluate $3x^2 - 4x$.

Substituting gives:

$$\begin{aligned} 3(-5)^2 - 4(-5) &= 3(25) + 20 && \text{using BIDMAS} \\ &= 75 + 20 \\ &= 95 \end{aligned}$$

2.2.8 Engineering Application

Engineering Application

Engineers frequently use formulae containing variables.

One of the most important examples is Newton's Second Law:

$$F = ma$$

where F is force, m is mass and a is acceleration.

If $m = 750$ kg and $a = 1.8$ m/s², then

$$F = (750)(1.8) = 1350 \text{ N}$$

Substitution allows engineers to use general equations to solve specific engineering problems.

2.2.9 Common Mistakes

Common errors include:

- Forgetting to expand brackets.

- Changing signs incorrectly when removing brackets.
- Forgetting brackets when substituting negative numbers.
- Ignoring BIDMAS after substitution.
- Assuming $2x^2 = (2x)^2$.
- Cancelling terms rather than factors in fractions.

2.2.10 Exercises

Exercise

1. Simplify:

(a) $3(x + 2) + 4x$

(b) $5(a - 3) + 2a$

(c) $4(2y - 1) + 3y$

(d) $2(3m + 4) - m$

2. Simplify:

(a) $x - (x + 5)$

(b) $4x - (2x - 3)$

(c) $6a - (3a + 4)$

(d) $5y - (2y - 7)$

3. Factorise:

(a) $4x + 8$

(b) $5y + 15$

(c) $6a + 9b$

(d) $12x^2 - 8xy$

4. Simplify:

(a) $\frac{8x}{4}$

(b) $\frac{15x^2}{5x}$

(c) $\frac{18ab}{6a}$

(d) $\frac{20x^3}{5x}$

5. Given $x = 3$, evaluate:

(a) $2x + 5$

(b) $4x - 7$

(c) $x^2 + 1$

Exercise

6. Given $a = 2$ and $b = 5$, evaluate:

(a) $3a + b$

(b) $2a^2 + b$

(c) $ab - 4$

7. Given $x = -2$, evaluate:

(a) x^2

(b) $3x + 4$

(c) $2x^2 - 5x$

Solutions to Exercise 2.2.10

1. (a) $7x + 6$

(b) $7a - 15$

(c) $11y - 4$

(d) $5m + 8$

2. (a) -5

(b) $2x + 3$

(c) $3a - 4$

(d) $3y + 7$

3. (a) $4(x + 2)$

(b) $5(y + 3)$

(c) $3(2a + 3b)$

(d) $4x(3x - 2y)$

4. (a) $2x$

(b) $3x$

(c) $3b$

(d) $4x^2$

5. (a) 11

(b) 5

(c) 10

6. (a) 11

(b) 13

(c) 6

7. (a) 4
(b) -2
(c) 18